

Mohammed Zuhair Hussain

AI Engineer @ Teclusion AI | Prev. AI Intern @ PwC | GenAI, LLMs & ML Systems

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Education

B.Tech, Artificial Intelligence Engineering — Mahindra University, Hyderabad

Sep 2022 – Aug 2026

CGPA **8.13/10** | Final-semester Industry Internship/Project graded **A+ (10/10)**

Finalist — Synchrony UniHack 2025 | Academic merit scholarship worth INR 100,000 for excellent academic performance

Experience

AI Engineer — Teclusion AI, Hyderabad, India

Jun 2026 – Present

- Building **Visentix**, a privacy-intelligence SaaS, as one of two engineers: an end-to-end LLM pipeline that decomposes legal text into clauses, classifies them across 8 privacy domains, and scores them deterministically (FastAPI, PostgreSQL/pgvector, self-hosted Qwen3-8B via Ollama), deployed on Azure, a RunPod GPU pod, and Cloudflare Workers.
- Keep the AI out of the arithmetic: 14 versioned deterministic formulas compute every score with full source-to-score lineage, SHA-256 regression-tested reproducible reports, and a guardrail filter that hard-fails any output containing banned legal-verdict language.
- Hardened the platform for production with row-level security across 56 Postgres tables, JWT authentication, per-endpoint rate limiting, retry/fallback logic for LLM calls, PII detection and redaction, and a **1,000+** test pytest suite.

AI/ML Engineer Intern — Teclusion AI, Hyderabad, India

Sep 2025 – Jun 2026

- Shipped **BluePearl**, an Intelligent Document Processing platform that lets financial teams ask one question across hundreds of reports and get accurate, source-cited answers — running entirely on locally hosted LLMs so client data never leaves their infrastructure, and cutting manual document review time by **90%**.
- Built **Vantage**, a Dynamic Pricing Engine combining ARIMA/SARIMA time-series forecasting with ML models to predict demand and price elasticity for every SKU, reacting to competitor moves within business-set margin and brand-price floors — driving a **6–10%** increase in optimized revenue and a **50%** reduction in manual pricing effort.

AI Intern, Emerging Tech — PricewaterhouseCoopers (PwC), Hyderabad, India

Jun 2025 – Aug 2025

- Developed a full-stack AI financial-advisory platform (FastAPI, Next.js, Firebase): an agentic system on Google Gemini that grounds personalised investment and tax guidance in a custom knowledge graph, answering in **under 2 seconds**.
- Co-authored an internal white paper (draft) on AI-enabled software development, driving research and framework design across LangChain, agentic AI, and SDLC optimisation with generative AI.
- Prototyped and demoed AI solutions for stakeholders, carrying features from concept through shipped implementation in an enterprise consulting environment.

Projects

Aperture — Cash-Flow Credit Decisioning Platform

[GitHub](#)

- Built end-to-end a credit-decisioning platform for applicants banks cannot score: underwrites from verified bank-transaction evidence (Account Aggregator) instead of a bureau score, served via REST APIs (FastAPI) with a React + TypeScript reviewer dashboard — a 9-gate deterministic policy engine makes every call, with zero AI in the decision path.
- Developed a transparent 11-feature cash-flow scorecard (scikit-learn) showing each feature's exact contribution to every decision, benchmarked against calibrated XGBoost — **ROC-AUC 0.787, KS 0.45** vs a 0.726 logistic baseline.
- Made decisions auditable and fair by design: point-in-time features with full lineage in PostgreSQL, a fairness allow-list blocking protected attributes at build time, 8 deterministic fraud detectors, and a hash-chained audit ledger.
- Added two grounded AI assistants ("explain this decision", "ask the codebase") that phrase only verified facts and fall back to templates rather than guess — backed by **393** automated tests and strict static typing.

Technologies: Python, FastAPI, PostgreSQL, scikit-learn, XGBoost, React, TypeScript, Docker on AWS EC2

Time Series Forecasting of Stock Prices — Model Benchmarking Study

[GitHub](#)

- Benchmarked five model families (ARIMA, GARCH, Prophet, KNN, feed-forward neural networks in TensorFlow) for 30-day stock-price forecasting, evaluating accuracy (RMSE, MAPE) vs. complexity trade-offs in a reproducible experiment pipeline.
- Grounded every comparison in rigorous statistical validation — ADF stationarity tests, ACF/PACF analysis, Ljung-Box diagnostics — with GARCH volatility modelling for risk-aware forecasts.

Technologies: Python, TensorFlow, Statsmodels, Prophet, scikit-learn, Pandas

WISP — Offline Mesh Messaging PWA

[GitHub](#)

- Architected a serverless Progressive Web App for crisis communication across three transport tiers (WebRTC over Wi-Fi, animated QR sync, printed posters), deployable via URL in **under 30 seconds** with zero backend — an offline-first architecture with IndexedDB persistence, TTL-based auto-cleanup, and a live force-directed mesh visualization.
- Engineered a gossip-style sync engine with Bloom-filter anti-entropy, Ed25519 signatures, and NaCl end-to-end encryption, achieving **sub-200 ms** LAN sync with chain-of-custody hop stamping and panic-wipe identity rotation, in a 180 KB gzipped bundle.

Technologies: React 18, TypeScript, WebRTC, IndexedDB (Dexie), Ed25519, TweetNaCl, Vite, Service Workers

Skills

Core Areas: Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, LLM Applications

Languages: Python, SQL, TypeScript/JavaScript, Java, C

ML & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, time-series forecasting (ARIMA, SARIMA, GARCH, Prophet), Statsmodels, NumPy, Pandas, OpenCV, model monitoring (calibration, drift)

LLM & GenAI: RAG, embeddings & vector search (pgvector, sentence-transformers), LangChain, LangGraph, Hugging Face, Ollama (local LLM serving), agentic systems, prompt engineering, LLM guardrails & evaluation

Backend & Web: FastAPI, REST APIs, async I/O, Node.js, Express.js, React, Next.js, Streamlit

Data & Cloud: PostgreSQL (Supabase, pgvector), MongoDB, Firebase, Docker, AWS (EC2), Azure, Vercel, Linux, Git/GitHub

Engineering Practices: multi-tenant security (RLS, JWT), rate limiting & fallbacks, pytest, mypy (strict), PII detection & redaction

Key Courses Taken

- **Machine Learning & AI:** Machine Learning and Python[#], Deep Neural Networks, Applied AI for Industry[#], Computational Sequence Modelling[#], Visual Data Analytics[#], Optimization Techniques for AI[#], Foundations of NLP, Artificial & Computational Intelligence, Digital Image Processing, Computational Genomics
- **Computer Science:** Data Structures, Design and Analysis of Algorithms, Object Oriented Programming[#], Operating Systems, Database Management Systems, Software Engineering, Computer Networks, Theory of Computing
- **Mathematics:** Mathematics I–III, Numerical Methods, Discrete Mathematical Structures, Signals & Systems

[#] Grade A or above

Positions of Responsibility

General Secretary — Mahindra University

Jun 2024 – Jun 2025

- Acted as the key liaison between **3,000+ students** and university management, resolving operational issues and driving collaborative problem-solving during events.
- Led a **team of 16** to organise 5 major campus events — Farewell, Orientation, Freshers, Convocation (for **1,000+ graduates**), and the cultural event “RRR” — owning planning, delegation, and on-ground execution.
- Expanded club activities and systemized the administrative structure, and contributed to the initial planning of **AEON**, the university’s first-ever Tech Fest.